

# Study guide for Exam AI-900: Microsoft Azure AI Fundamentals

## Purpose of this document

This study guide should help you understand what to expect on the exam and includes a summary of the topics the exam might cover and links to additional resources. The information and materials in this document should help you focus your studies as you prepare for the exam.

 Expand table

| Useful links                                    | Description                                                                                                                                      |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">How to earn the certification</a>   | Some certifications only require passing one exam, while others require passing multiple exams.                                                  |
| <a href="#">Certification renewal</a>           | Microsoft associate, expert, and specialty certifications expire annually. You can renew by passing a free online assessment on Microsoft Learn. |
| <a href="#">Your Microsoft Learn profile</a>    | Connecting your certification profile to Microsoft Learn allows you to schedule and renew exams and share and print certificates.                |
| <a href="#">Exam scoring and score reports</a>  | A score of 700 or greater is required to pass.                                                                                                   |
| <a href="#">Exam sandbox</a>                    | You can explore the exam environment by visiting our exam sandbox.                                                                               |
| <a href="#">Request accommodations</a>          | If you use assistive devices, require extra time, or need modification to any part of the exam experience, you can request an accommodation.     |
| <a href="#">Take a free Practice Assessment</a> | Test your skills with practice questions to help you prepare for the exam.                                                                       |

## Updates to the exam

Our exams are updated periodically to reflect skills that are required to perform a role. We have included two versions of the Skills Measured objectives depending on when you are taking the exam.

We always update the English language version of the exam first. Some exams are localized into other languages, and those are updated approximately eight weeks after the English version is

updated. While Microsoft makes every effort to update localized versions as noted, there may be times when the localized versions of an exam are not updated on this schedule. Other available languages are listed in the **Schedule Exam** section of the **Exam Details** webpage. If the exam isn't available in your preferred language, you can request an additional 30 minutes to complete the exam.

## Note

The bullets that follow each of the skills measured are intended to illustrate how we are assessing that skill. Related topics may be covered in the exam.

## Note

Most questions cover features that are general availability (GA). The exam may contain questions on Preview features if those features are commonly used.

# Skills measured as of May 2, 2025

## Audience profile

This exam is an opportunity for you to demonstrate knowledge of machine learning and AI concepts and related Microsoft Azure services. As a candidate for this exam, you should have familiarity with Exam AI-900's self-paced or instructor-led learning material.

This exam is intended for you if you have both technical and non-technical backgrounds. Data science and software engineering experience are not required. However, you would benefit from having awareness of:

- Basic cloud concepts
- Client-server applications

You can use Azure AI Fundamentals to prepare for other Azure role-based certifications like Azure Data Scientist Associate or Azure AI Engineer Associate, but it's not a prerequisite for any of them.

## Skills at a glance

- Describe Artificial Intelligence workloads and considerations (15–20%)

- Describe fundamental principles of machine learning on Azure (15–20%)
- Describe features of computer vision workloads on Azure (15–20%)
- Describe features of Natural Language Processing (NLP) workloads on Azure (15–20%)
- Describe features of generative AI workloads on Azure (20–25%)

## **Describe Artificial Intelligence workloads and considerations (15–20%)**

### **Identify features of common AI workloads**

- Identify computer vision workloads
- Identify natural language processing workloads
- Identify document processing workloads
- Identify features of generative AI workloads

### **Identify guiding principles for responsible AI**

- Describe considerations for fairness in an AI solution
- Describe considerations for reliability and safety in an AI solution
- Describe considerations for privacy and security in an AI solution
- Describe considerations for inclusiveness in an AI solution
- Describe considerations for transparency in an AI solution
- Describe considerations for accountability in an AI solution

## **Describe fundamental principles of machine learning on Azure (15–20%)**

### **Identify common machine learning techniques**

- Identify regression machine learning scenarios

- Identify classification machine learning scenarios
- Identify clustering machine learning scenarios
- Identify features of deep learning techniques
- Identify features of the Transformer architecture

## **Describe core machine learning concepts**

- Identify features and labels in a dataset for machine learning
- Describe how training and validation datasets are used in machine learning

## **Describe Azure Machine Learning capabilities**

- Describe capabilities of automated machine learning
- Describe data and compute services for data science and machine learning
- Describe model management and deployment capabilities in Azure Machine Learning

## **Describe features of computer vision workloads on Azure (15–20%)**

### **Identify common types of computer vision solution**

- Identify features of image classification solutions
- Identify features of object detection solutions
- Identify features of optical character recognition solutions
- Identify features of facial detection and facial analysis solutions

### **Identify Azure tools and services for computer vision tasks**

- Describe capabilities of the Azure AI Vision service
- Describe capabilities of the Azure AI Face detection service

## **Describe features of Natural Language Processing (NLP) workloads on Azure (15–20%)**

### **Identify features of common NLP Workload Scenarios**

- Identify features and uses for key phrase extraction
- Identify features and uses for entity recognition
- Identify features and uses for sentiment analysis
- Identify features and uses for language modeling
- Identify features and uses for speech recognition and synthesis
- Identify features and uses for translation

### **Identify Azure tools and services for NLP workloads**

- Describe capabilities of the Azure AI Language service
- Describe capabilities of the Azure AI Speech service

## **Describe features of generative AI workloads on Azure (20–25%)**

### **Identify features of generative AI solutions**

- Identify features of generative AI models
- Identify common scenarios for generative AI
- Identify responsible AI considerations for generative AI

### **Identify generative AI services and capabilities in Microsoft Azure**

- Describe features and capabilities of Azure AI Foundry
- Describe features and capabilities of Azure OpenAI service

- Describe features and capabilities of Azure AI Foundry model catalog

## Study resources

We recommend that you train and get hands-on experience before you take the exam. We offer self-study options and classroom training as well as links to documentation, community sites, and videos.

 Expand table

| Study resources        | Links to learning and documentation                                                                                                                                                                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Get trained            | <a href="#">Choose from self-paced learning paths and modules or take an instructor-led course</a>                                                                                                                                                                                                                             |
| Find documentation     | <a href="#">Anomaly Detector</a><br><a href="#">Language Understanding</a><br><a href="#">Azure Machine Learning</a><br><a href="#">Computer Vision</a><br><a href="#">Natural language processing technology</a><br><a href="#">Azure Bot Service</a><br><a href="#">Speech to Text</a><br><a href="#">Speech Translation</a> |
| Ask a question         | <a href="#">Microsoft Q&amp;A   Microsoft Docs</a>                                                                                                                                                                                                                                                                             |
| Get community support  | <a href="#">Artificial Intelligence and Machine Learning Hub</a>                                                                                                                                                                                                                                                               |
| Follow Microsoft Learn | <a href="#">Microsoft Learn - Microsoft Tech Community</a>                                                                                                                                                                                                                                                                     |
| Find a video           | <a href="#">The AI Show</a><br><a href="#">Browse other Microsoft Learn shows</a>                                                                                                                                                                                                                                              |

## Change log

The table below summarizes the changes between the current and previous version of the skills measured. The functional groups are in bold typeface followed by the objectives within each group. The table is a comparison between the previous and current version of the exam skills measured and the third column describes the extent of the changes.

 Expand table

| Skill area prior to May 2, 2025                               | Skill area as of May 2, 2025                                        | Change                  |
|---------------------------------------------------------------|---------------------------------------------------------------------|-------------------------|
| Describe Artificial Intelligence workloads and considerations | Describe Artificial Intelligence workloads and considerations       | No change               |
| Identify features of common AI workloads                      | Identify features of common AI workloads                            | Major                   |
| Describe fundamental principles of machine learning on Azure  | Describe fundamental principles of machine learning on Azure        | % of the exam decreased |
| Identify common machine learning techniques                   | Identify common machine learning techniques                         | Minor                   |
| Describe features of generative AI workloads on Azure         | Describe features of generative AI workloads on Azure               | % of the exam increased |
| Identify capabilities of Azure OpenAI Service                 | Identify generative AI services and capabilities in Microsoft Azure | Major                   |

Last updated on 05/05/2025